

PROJECT DEVELOPMENT PHASE

1. Dataset Collection

The cosmetics dataset used in this project was collected from the Kaggle platform. The dataset contains information about cosmetic products, brands, prices, ingredients, rankings, and skin types.

Label	Brand	Name	Price	Rank	Ingredients	Combination Dry	Normal	Oily	Sensitive
1	Moisturize LA MER	Crème de la Mer	175	4.1	Algae (Seaweed) Extract, Min	1	1	1	1
2	Moisturize SK-II	Facial Treatment Essence	179	4.1	Galactomyces Ferment Filtrat	1	1	1	1
3	Moisturize DRUNK ELEPHANT	Protini™ Polypeptide Cream	68	4.4	Water, Dicaprylyl Carbonate, i	1	1	1	0
4	Moisturize LA MER	The Moisturizing Soft Cream	175	3.8	Algae (Seaweed) Extract, Cycl	1	1	1	1
5	Moisturize IT COSMETICS	Your Skin But Better™ CC+™ Cream with SPF 50+	38	4.1	Water, Snail Secretion Filtrate	1	1	1	1
6	Moisturize TATCHA	The Water Cream	68	4.2	Water, Saccharomyces/Came	1	0	1	1
7	Moisturize DRUNK ELEPHANT	Lala Retro™ Whipped Cream	60	4.2	Water, Glycerin, Caprylic/ Cap	1	1	1	0
8	Moisturize DRUNK ELEPHANT	Virgin Marula Luxury Facial Oil	72	4.4	100% Unrefined Sclerocarya E	1	1	1	0
9	Moisturize KIEHL'S SINCE 1851	Ultra Facial Cream	29	4.4	Water, Glycerin, Cyclohexasilc	1	1	1	1
10	Moisturize LA MER	Little Miss Miracle Limited-Edition Crème de la Mer	325	5	Algae (Seaweed) Extract, Mine	0	0	0	0
11	Moisturize FRESH	Lotus Youth Preserve Moisturizer	45	4.3	Water, Glycerin, Propylene Gly	0	0	0	0
12	Moisturize KIEHL'S SINCE 1851	Midnight Recovery Concentrate	47	4.4	Caprylic/Capric Triglyceride Di	1	1	1	1
13	Moisturize BELIF	The True Cream Aqua Bomb	38	4.5	Water, Dipropylene Glycol, Gh	1	0	1	0
14	Moisturize SUNDAY RILEY	Luna Sleeping Night Oil	105	4.1	Persea Grattissima (Extra Virgi	1	1	1	1
15	Moisturize FARMACY	Honeymoon Glow AHA Resurfacing Night Serum with Echinacea	58	4.6	Water, Lactic Acid, Propanedi	1	1	1	1
16	Moisturize DRUNK ELEPHANT	The Little™	90	4.4	Beste™ No.9 Jelly Cleanser: W	1	1	1	0
17	Moisturize FIRST AID BEAUTY	Ultra Repair™ Cream Intense Hydration	30	4.6	Water, Stearic Acid, Glycerin, i	1	1	1	1
18	Moisturize CLINIQUE	Moisture Surge 72-Hour Auto-Replenishing Hydrator	39	4.4	Water, Dimethicone, Butylen	1	1	1	1
19	Moisturize FRESH	Rose Deep Hydration Moisturizer	40	4.4	Water, Glycerin, Ethylhexylso	0	0	0	0
20	Moisturize SK-II	R.N.A. POWER Face Cream	230	4.3	Water, Glycerin, Galactomyce	0	1	1	0
21	Moisturize LA MER	Crème de la Mer Mini	85	4.1	Algae (Seaweed) Extract, Mine	1	1	1	1
22	Moisturize BAREMINERALS	COMPLEXION RESCUE™ Tinted Moisturizer Broad Spectrum SPF	30	3.9	Water, Coconut Alkanes, Prop	0	0	0	0
23	Moisturize SHISEIDO	Bio-Performance Advanced Super Revitalizing Cream	78	4.6	Water, Glycerin, Cyclomethicc	0	0	0	0
24	Moisturize FRESH	Black Tea Firming Overnight Mask	92	4.1	Water, Glycerin, Butylene Glyc	1	1	1	0
25	Moisturize BELIF	The True Cream Moisturizing Bomb	38	4.6	Water, Glycerin, Cyclohexasilc	0	1	1	0
26	Moisturize CHARLOTTE TILBURY	Charlotte's Magic Cream	100	4.4	Water, Homosalate, Glyceryl	0	0	0	0
27	Moisturize DRUNK ELEPHANT	Virgin Marula Luxury Facial Oil Mini	40	4.5	100% Unrefined Sclerocarya E	1	1	1	0
28	Moisturize ORIGINS	Dr. Andrew Weil For Origins™ Mega-Mushroom Relief & Resilien	34	4.4	Water, Butylene Glycol, PEG-4	1	1	1	1
29	Moisturize CLINIQUE	Dramatically Different Moisturizing Lotion+	28	3.9	Water, Mineral Oil/Paraffinur	1	1	0	0
30	Moisturize SK-II	GenOptics Aura Essence Serum	240	4.1	Water, Galactomyces Ferment	1	1	1	1

2. Data Preprocessing Steps

The following preprocessing steps were performed before creating visualizations:

Collected the cosmetics dataset from Kaggle.

Imported the dataset into Microsoft Excel and Tableau.

Performed data cleaning and validation.

Checked for missing values and duplicate records.

Organized the dataset into meaningful categories.

Prepared the dataset for visualization and analysis.

Imported the processed dataset into Tableau.

3. Business Questions

The following business questions were identified and analyzed:

Which cosmetic categories contain the highest number of products?

What is the average price distribution among different cosmetic brands?

Which are the top cosmetic brands available in the dataset?

Which cosmetic products are suitable for different skin types?

Which cosmetic categories generate higher revenue?

What is the relationship between product price and ranking?

Which brands have the highest product count?

What are the major customer preferences in cosmetic products?

4. Visualizations

Chart 1: Category-wise Distribution of Cosmetic Products

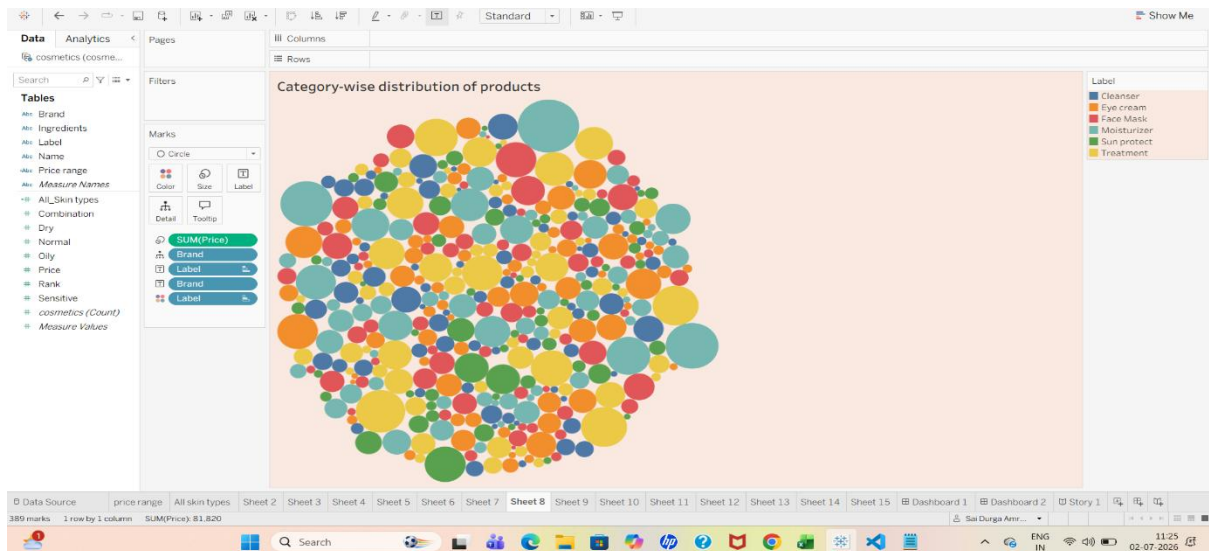


Chart 2: Brand-wise Average Price Analysis

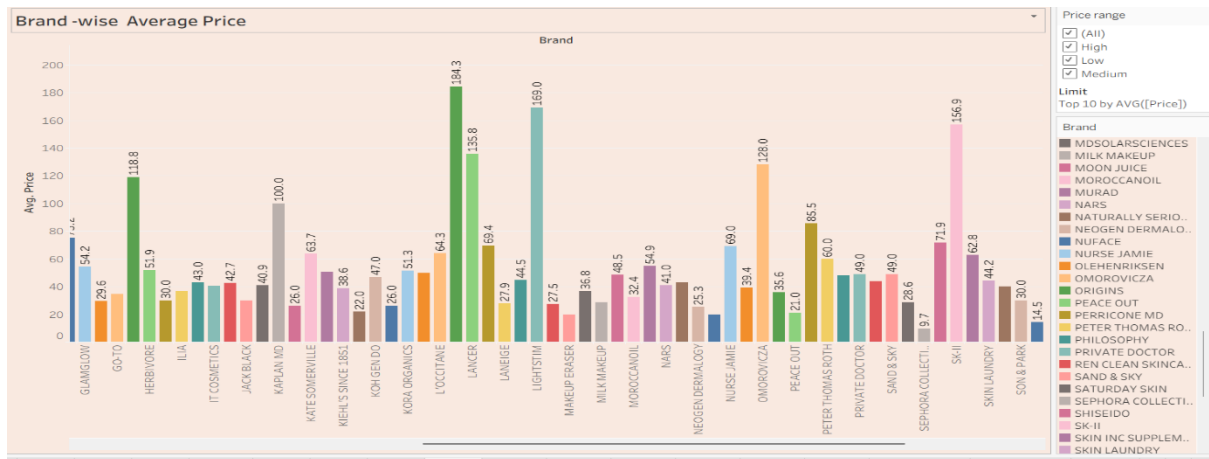


Chart 3: Top 10 Cosmetic Brands Analysis

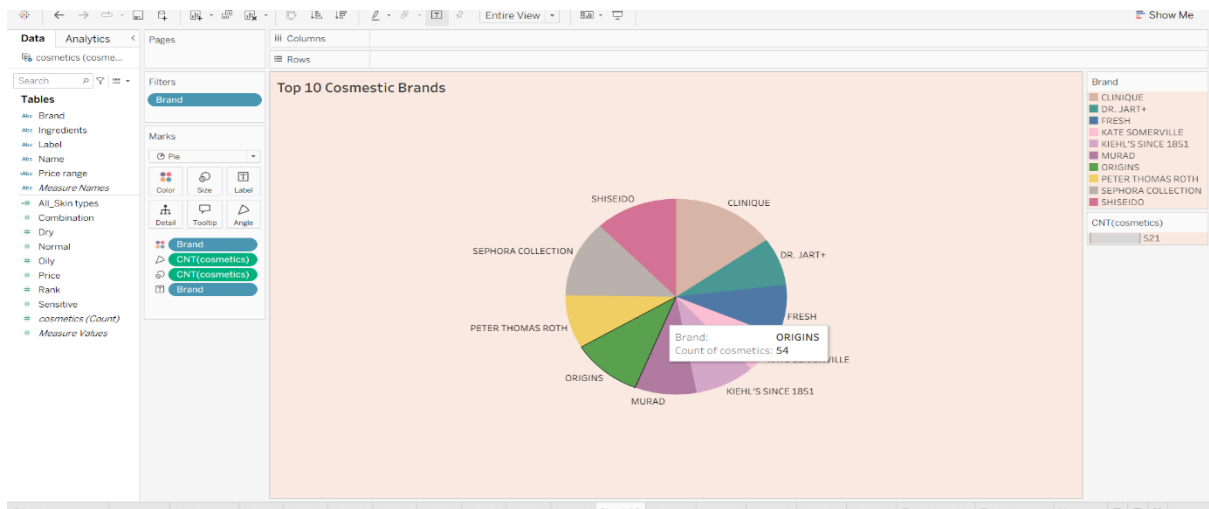


Chart 4: Category-wise Distribution of Combination Skin Products



Chart 5: Top 10 Brands Suitable for All Skin Types

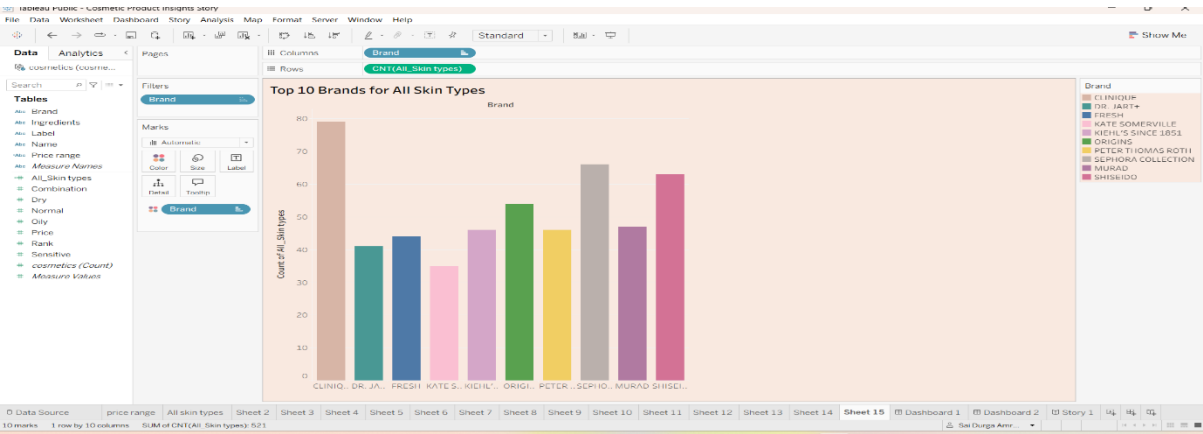


Chart 6: Category-wise Total Revenue Analysis

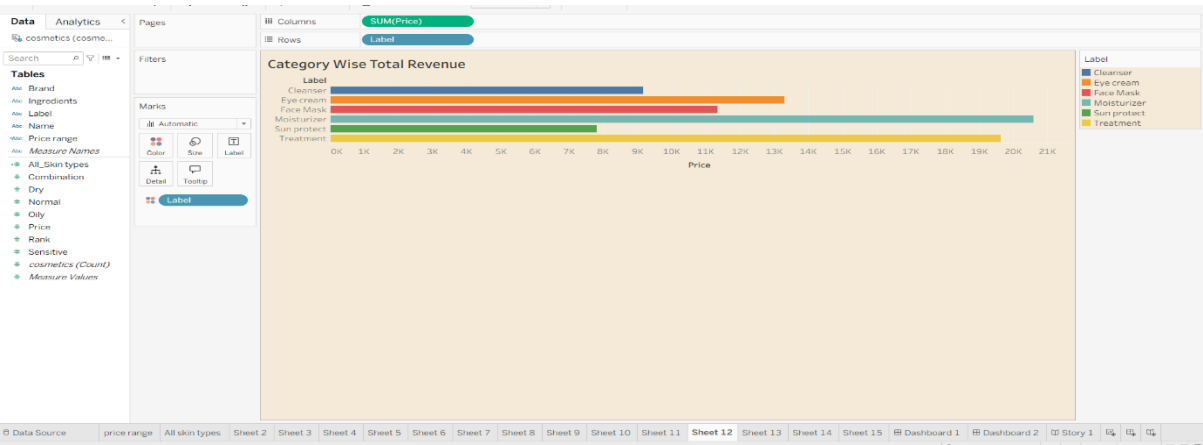


Chart 7: Top 10 Products by Product Count

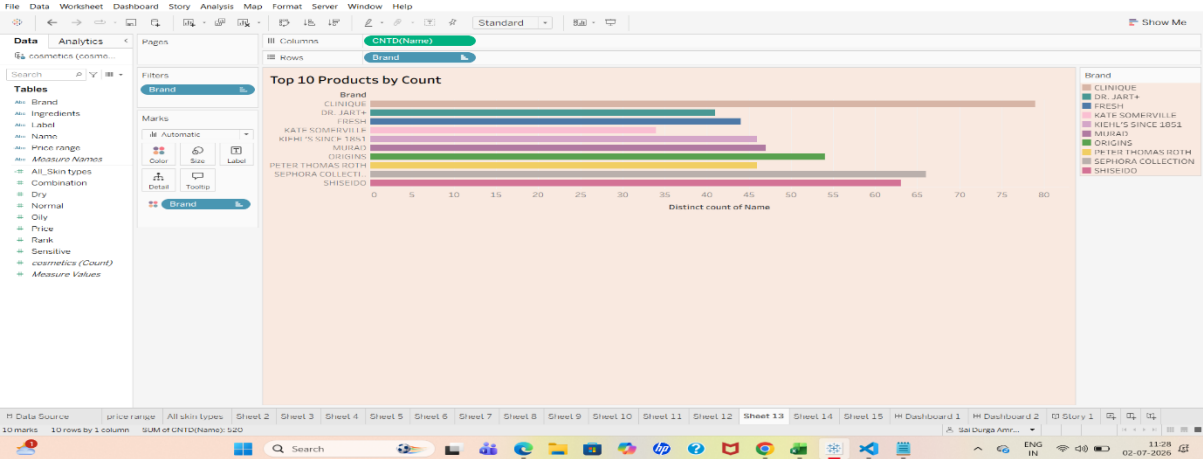


Chart 8: Price vs Rank Analysis using Scatter Plot

